

ATRIK SAMANTA

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EDUCATION

B.Tech in Computer Science and Engineering (AI & ML) Vellore Institute of Technology (VIT), Vellore CGPA: 7.9	2023 - 2027
Class XII (CBSE) Hem Sheela Model School	2023
Class X (CBSE) Hem Sheela Model School	2021

SKILLS AND INTERESTS

Languages	Python, R Language, SQL, Java, C/C++, JavaScript, HTML/CSS
AI Tools & Agents	Claude Code, Model Context Protocol (MCP), GitHub Copilot, Cursor, LLMs (Llama, Qwen)
Optimization & ML	Google OR-Tools (Linear Programming, VRPTW), PyTorch, TensorFlow, Scikit-learn
Frameworks & Web	FastAPI, React, Next.js, Node.js, Streamlit, Vite
Cloud, DevOps & DB	AWS (EC2), Docker, Git, Linux, Nginx, PostgreSQL, SQLite, Supabase
Data & Analytics	NumPy, pandas, OpenCV, Matplotlib, Data Quality Assessment, Lineage Tracking
Core CS	Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming
Soft Skills	Stakeholder Communication, Process Automation, Cross-Functional Teamwork, Problem Solving
Spoken Languages	English, Hindi, Bengali, German (Level 1)

PROJECTS

Vehicle Routing Problem with Time Windows (VRPTW) July 2026 – August 2026
Python, Google OR-Tools, SQLite, Streamlit, Folium, Plotly, OSRM

- Engineered an automated VRPTW optimization engine using Google OR-Tools to solve complex multi-constraint logistics workflows under vehicle capacity, customer time-windows, and route duration bounds.
- Prototyped interactive Streamlit analytics dashboards using Claude Code and Python for scenario testing, route mapping, and Plotly Gantt schedule tracking to measure delivery KPIs.
- Curated relational SQLite data schemas and integrated OSRM network matrices with disjunction-based infeasibility handling to eliminate routing bottlenecks across transport networks.

AI-Based Logistics Network Optimizer June 2026 – July 2026
Python, Google OR-Tools, SQLite, Streamlit, Plotly, NetworkX, Machine Learning

- Developed a supply chain network optimization system using Linear Programming (OR-Tools) to automate transportation allocation and warehouse capacity management.
- Defined structured SQLite backend schemas to enforce data quality thresholds, track shipment data lineage, and maintain reliable data inputs across supply chain domains.
- Integrated ML-based demand forecasting models (Random Forest/XGBoost) into the optimization pipeline to enhance predictive planning accuracy and quantify operational productivity gains.

Intrusion Detection System for IoT Networks

July 2026 - Present

Personal Project

- Developing an AI-driven threat classification pipeline leveraging PyTorch and Scikit-learn to automate real-time cyber threat detection and anomaly mitigation across IoT network traffic.
- Processed and evaluated high-dimensional network telemetry datasets, establishing data quality baselines to identify malicious traffic patterns and Denial-of-Service (DoS) attacks.
- Optimized feature engineering and model architecture to deliver high classification accuracy with minimal processing latency for resource-constrained edge environments.

StructuralEye: Crack Detection System

March 2026 - April 2026

Personal Project

- Automated manual structural inspection workflows using computer vision algorithms including Canny edge detection, CLAHE, and morphological operations.
- Extracted quantitative physical dimensions (length, width, area) to compute dynamic risk scores (0–100) and establish baseline severity KPIs for defect monitoring.
- Visualized structural degradation trends over time via an interactive analytics dashboard to support data-driven maintenance decisions.

AI-Based Fake News Detection System

December 2025 - January 2026

Personal Project

- Built an NLP classification pipeline using TF-IDF feature extraction and text preprocessing to classify content and automate information verification workflows.
- Evaluated multiple machine learning models (Logistic Regression, Naive Bayes) against precision, recall, and F1-score metrics to ensure reliable prediction accuracy.

CERTIFICATIONS

IBM Advanced Generative AI Certification, IBM Career Education

May 2026 - July 2026

Core Focus: Foundation Models, Transformers, LLMs, GANs, Diffusion Models, Advanced Prompt Engineering (Chain-of-Thought, Tree-of-Thoughts), and Agentic Deployment via IBM watsonx.ai, IBM Granite, Hugging Face, and GitHub Copilot.

POSITIONS OF RESPONSIBILITY

Events Head

2024 - 2025

The Otaku Club

- Formulated, structured, and executed end-to-end interactive student events and operational workflows.
- Led cross-functional student teams across logistics, marketing, and operations to align stakeholder requirements and drive initiative execution.

ACHIEVEMENTS

- Built and deployed end-to-end AI/ML systems focused on Operations Research, Supply Chain Optimization, Computer Vision, and Natural Language Processing.
- Elected and served as **Events Head** of The Otaku Club for the 2024–2025 academic year, managing event logistics and team delivery.